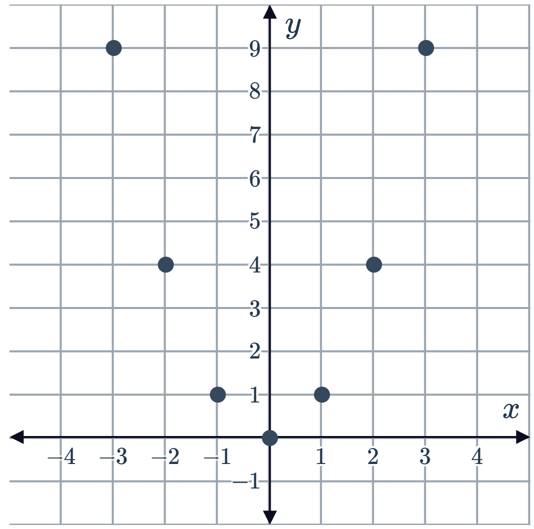
Graphing quadratic functions

1

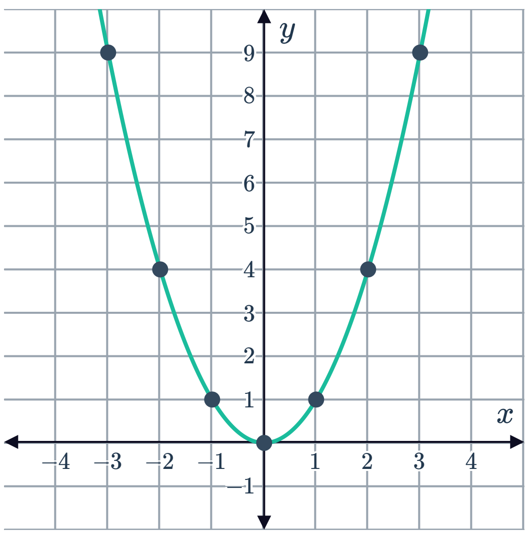
a

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

b



c



d No

e $x = 0$

f 0

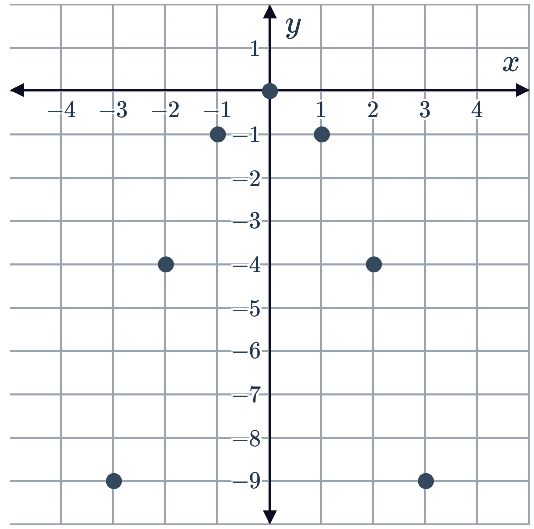
g 2

2

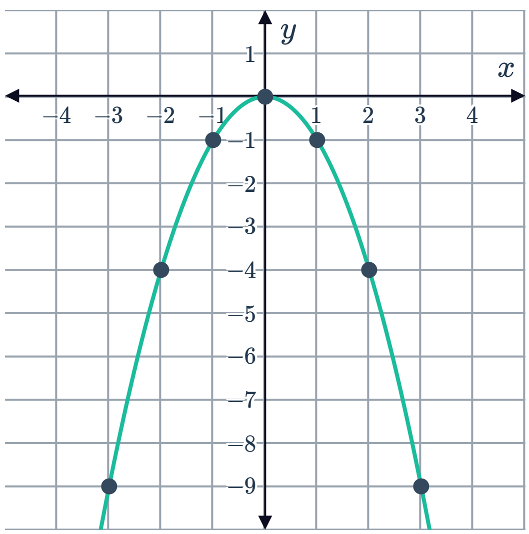
a

x	-3	-2	-1	0	1	2	3
y	-9	-4	-1	0	-1	-4	-9

b



c



d No

e 0

f $x = 0$

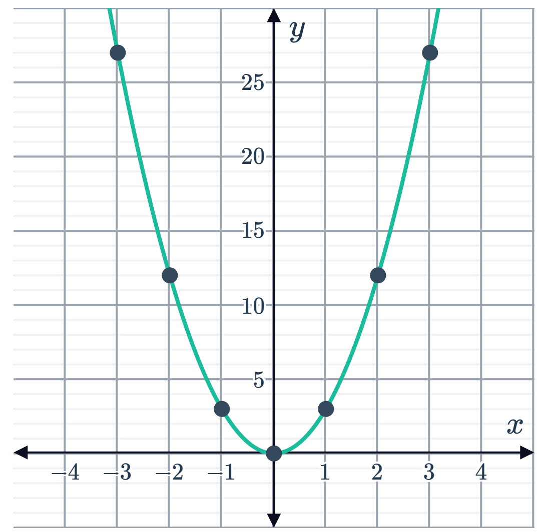
3

a

i

x	-3	-2	-1	0	1	2	3
y	27	12	3	0	3	12	27

ii

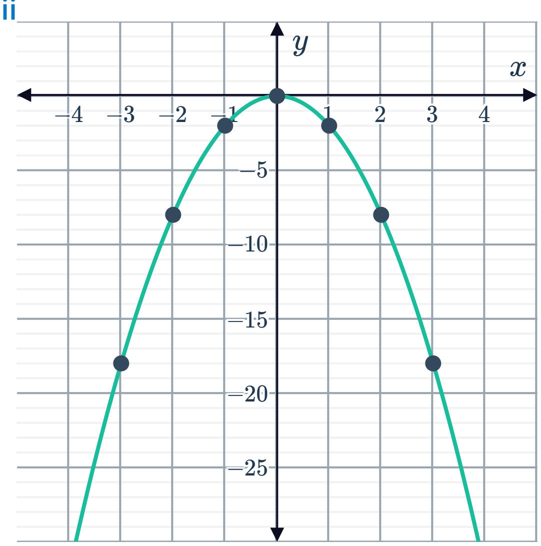


b

i

x	-3	-2	-1	0	1	2	3
y	-18	-8	-2	0	-2	-8	-18

ii

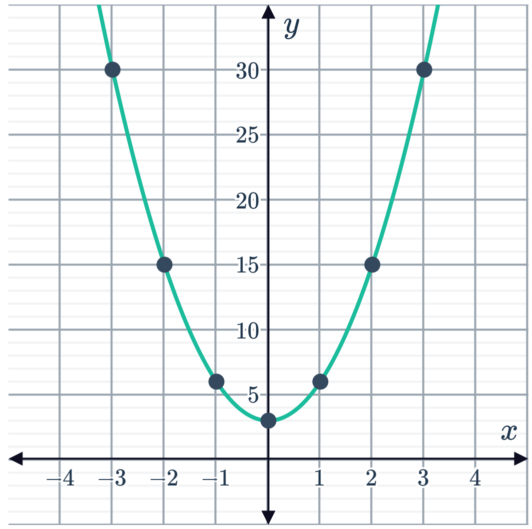


c

i

x	-3	-2	-1	0	1	2	3
y	30	15	6	3	6	15	30

ii

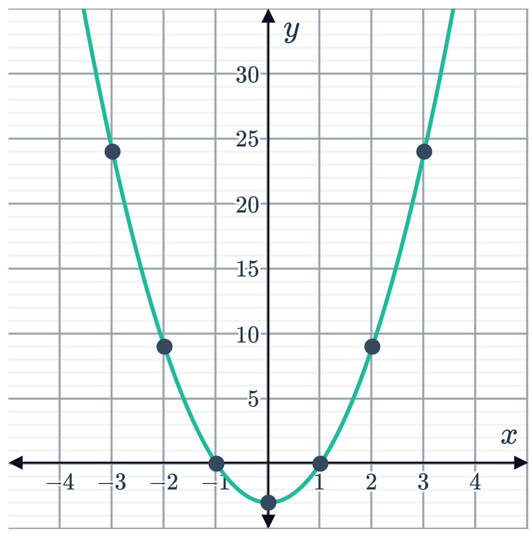


d

i

x	-3	-2	-1	0	1	2	3
y	24	9	0	-3	0	9	24

ii

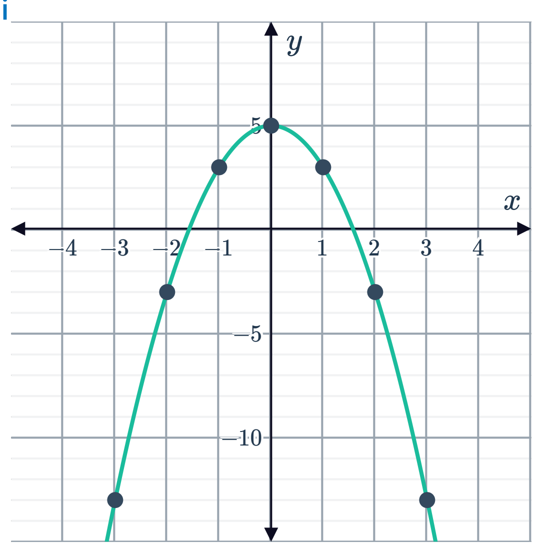


e

i

x	-3	-2	-1	0	1	2	3
y	-13	-3	3	5	3	-3	-13

ii

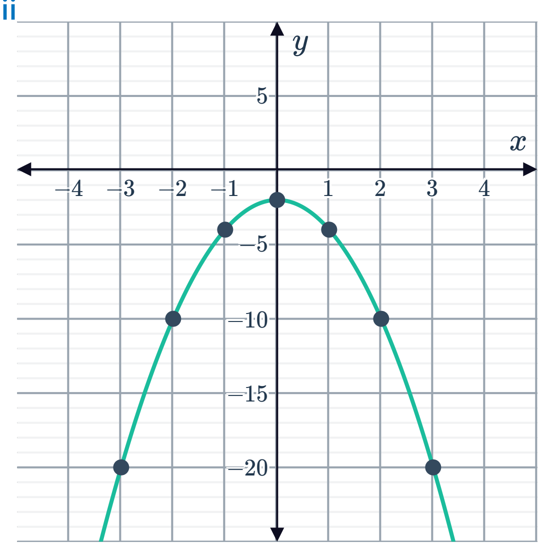


f

i

x	-3	-2	-1	0	1	2	3
y	-20	-10	-4	-2	-4	-10	-20

ii



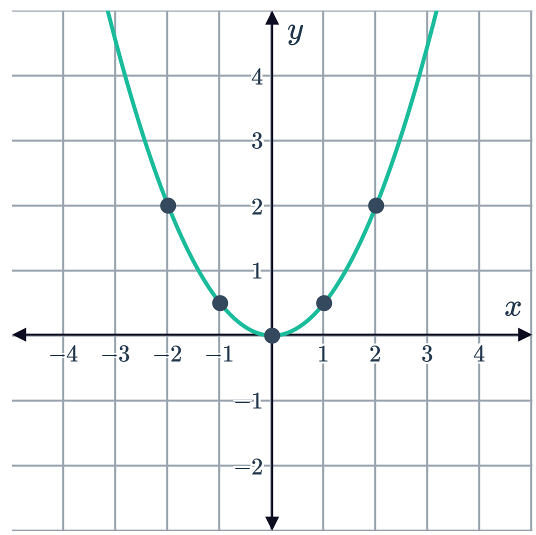
4

a

i

x	-2	-1	0	1	2
y	2	$\frac{1}{2}$	0	$\frac{1}{2}$	2

ii

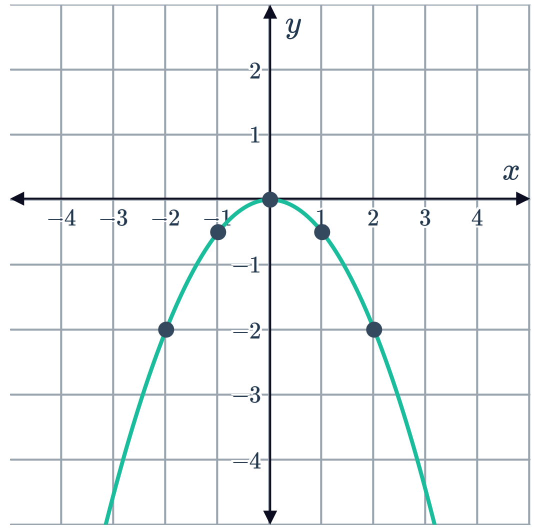


b

i

x	-2	-1	0	1	2
y	-2	$-\frac{1}{2}$	0	$-\frac{1}{2}$	-2

ii

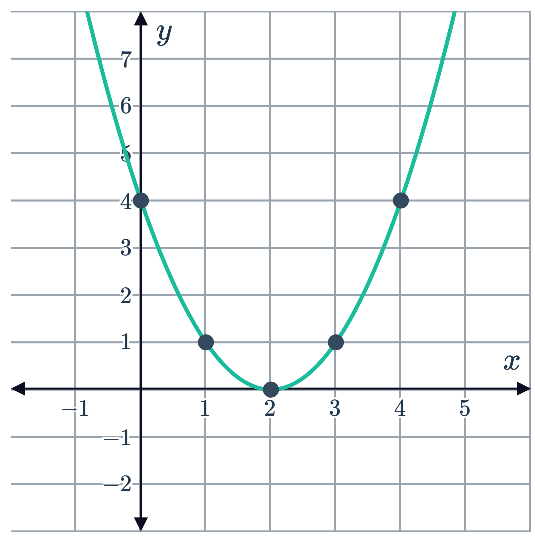


5

a

x	0	1	2	3	4
y	4	1	0	1	4

b



c 0

d 2

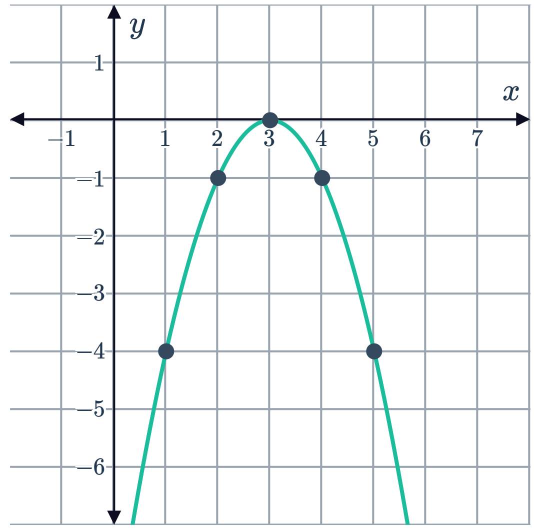
e $(2, 0)$

6

a

x	1	2	3	4	5
y	-4	-1	0	-1	-4

b



c 0

d 3

e $(3, 0)$

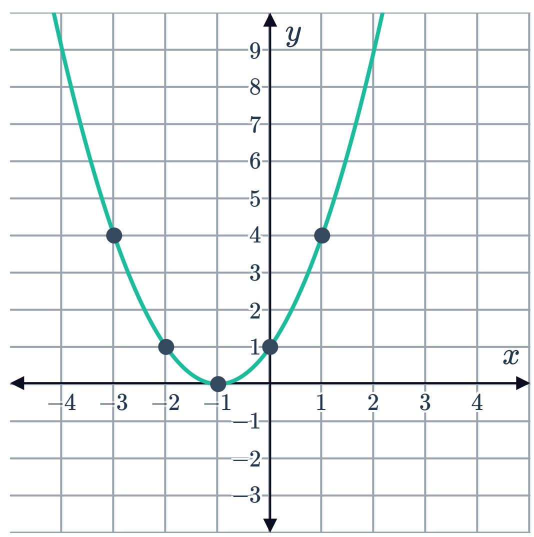
7

a

i

x	-3	-2	-1	0	1
y	4	1	0	1	4

ii



iii 0

iv -1

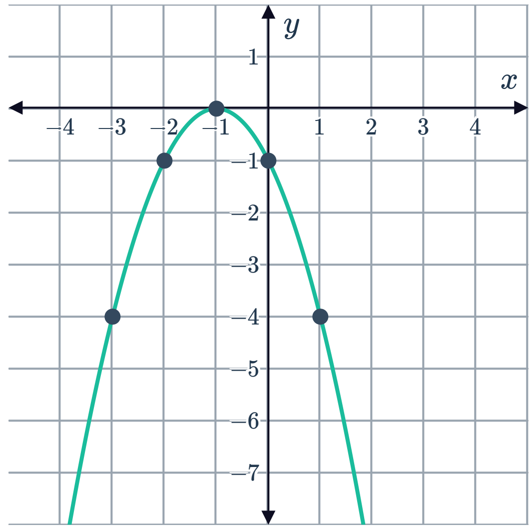
v $(-1, 0)$

b

i

x	-3	-2	-1	0	1
y	-4	-1	0	-1	-4

ii



iii 0

iv -1

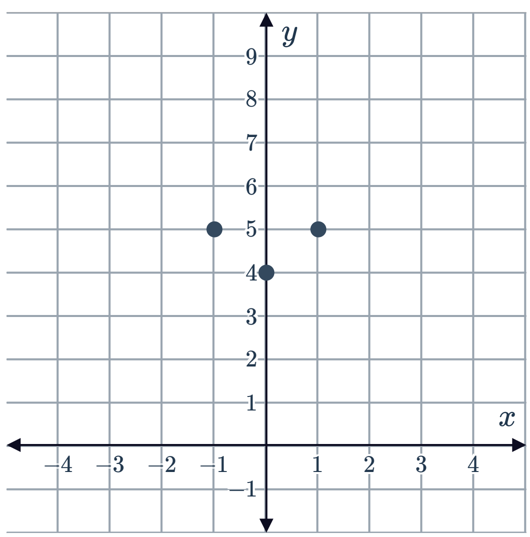
v $(-1, 0)$

8 True

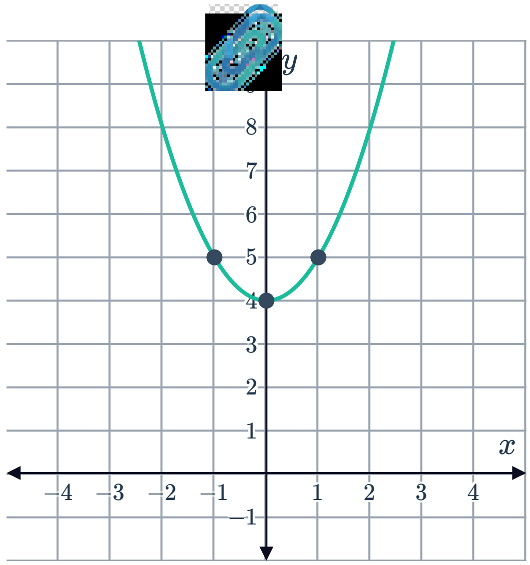
9

a $A(-3, 13)$, $B(-2, 8)$, $C(-1, 5)$, $D(0, 4)$, $E(1, 5)$, $F(2, 8)$, $G(3, 13)$

b



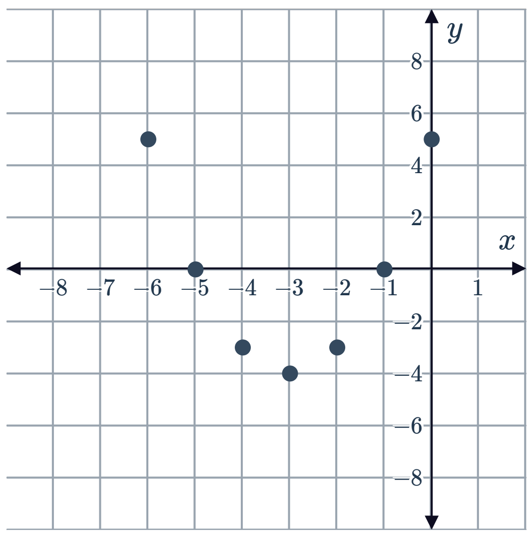
c



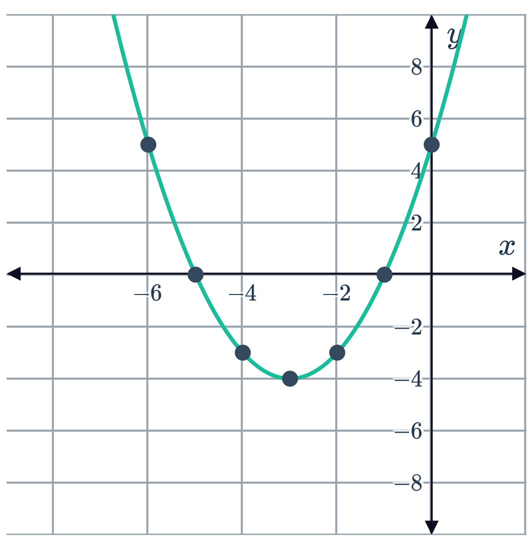
10

a $A(-5,0)$, $B(-2,-3)$, $C(-6,5)$, $D(-1,0)$, $E(0,5)$, $F(-3,-4)$, $G(-4,-3)$

b



c



11 $y = 0$

12 Maximum value

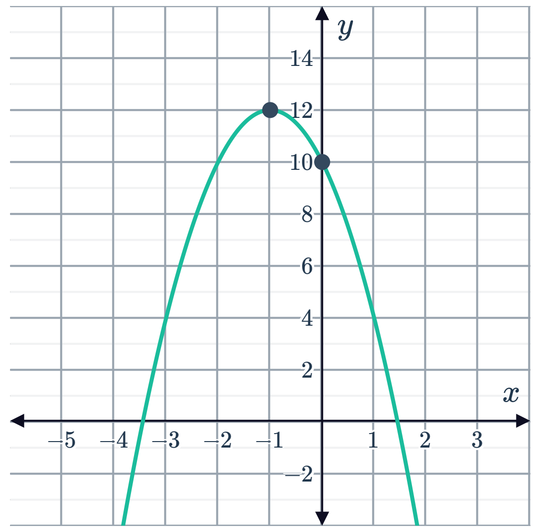
13

a $x = -1$

b 12

c $(-1, 12)$

d

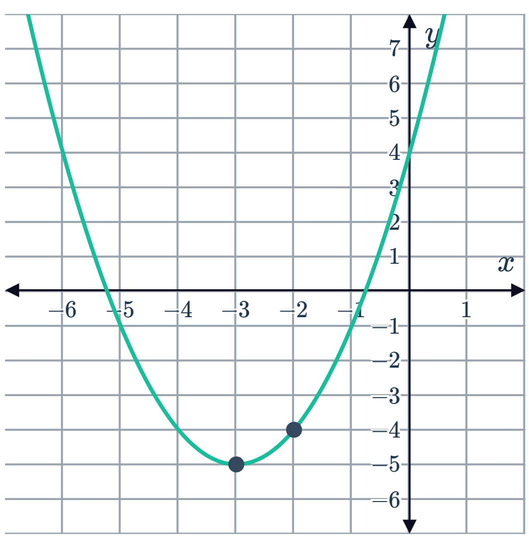


14

a $x = -3$

b -5

c



15

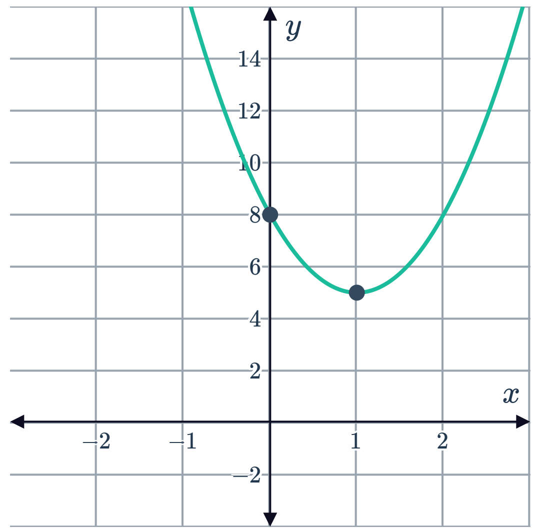
a

i $x=1$

ii $(1,5)$

iii 8

iv



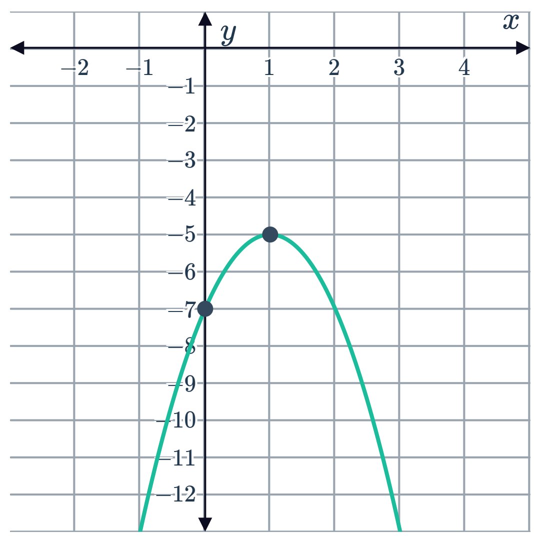
b

i $x=1$

ii $(1,-5)$

iii -7

iv

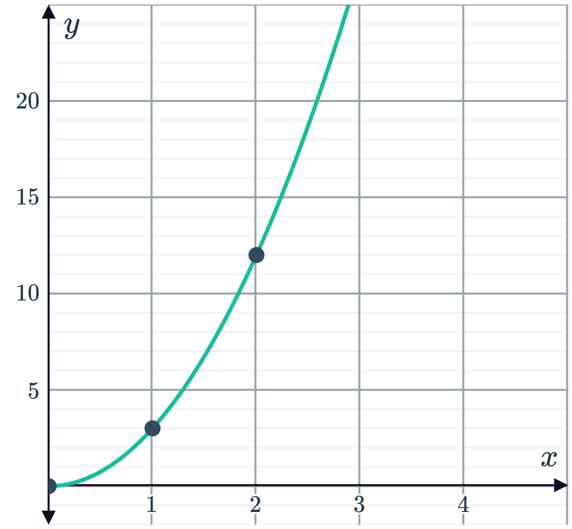


16

a

x (m)	0	1	2	3	4
y (m)	0	3	12	27	48

b



c $0\ \mathrm{m}$

d 0

e $(0, 0)$

17 $g\left(x\right)$ has the smallest minimum value which is -16